

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov – Dec 2015

Max. Marks: 100

Class: SE

Name of the Course: Discrete Structures (DIS)

Course Code: UCEC305

Duration: 3Hrs

Semester: III

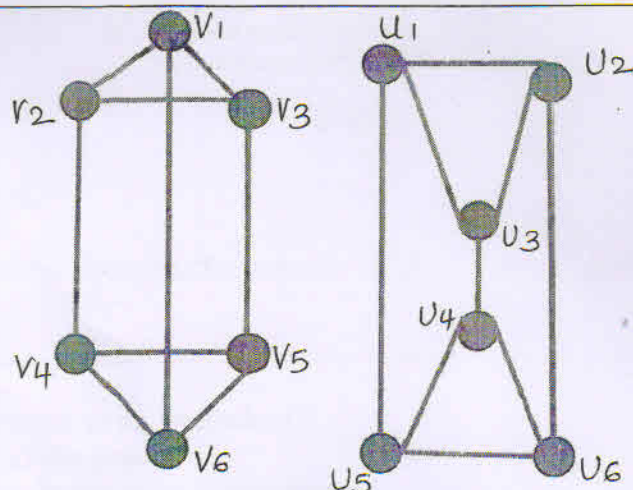
Branch: Comp

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Prove by mathematical induction : $2 + 5 + 8 + \dots + (3n - 1) = n(3n + 1)/2$	05
Q 1 (b)	Determine the number of positive integers n where $1 \leq n \leq 100$ and n is not divisible by 2, 3 or 5.	05
Q 1 (c)	Consider the equilateral triangle of side of one unit each. Show that if any four points are chosen in the region, then two of them must be no further apart than one unit	05
Q1 (d)	A relation R on A is called circular if aRb and bRc imply cRa . Show that R is reflexive and circular if and only if it is an equivalence relation.	05
Q2 (a)	Find the Hasse diagram of D_{36} . Check whether it is a lattice and find two sublattices.	10
Q2 (b)	(i) Let $A = \{1, 4, 7, 13\}$ and relation $R = \{(1, 4), (4, 7), (7, 4), (1, 13)\}$ Find the transitive closure of R using Warshall's Algorithm. (ii) Let $f(x) = x^3$, $g(x) = 4x^2 + 1$, $h(x) = 7x - 2$. Find $f \circ h \circ g$ and $g \circ h \circ f$	10
OR		
Q2 (a)	Draw Hasse Diagram of D_{42} . Check whether it is a Boolean algebra ?	10

Q2 (b)	Let $A = \{1, 2, 3, 4, 6, 12, 24\}$ Find R such that aRb iff $a \leq b$. (iii) Compute R^2, MR^2, draw digraph. (iv) Compute R^3, MR^3, draw digraph (v) Define Connectivity relation and find out it for the above relation.	10
Q 3(a)	(i) Solve the recurrence relation for $a_n - 2a_{n-1} + 2a_{n-2} - a_{n-3} = 0$ (ii) Find out the generating functions for $1, 1, 1, 1, 1, 1, \dots$ $1, 2, 3, 4, 5, \dots$	10
Q3 (b)	(i) Using laws of logic prove that the following statement is a tautology. $[(P \rightarrow (P \wedge Q)) \rightarrow \sim Q]$ is a Tautology (ii) Show that if 30 dictionaries in a library contain a total of 61,337 pages, then one of the dictionaries must have at least 2045 pages.	10
OR		
Q 3(b)	(i) Explain universal and existential quantifiers, tautology with examples.(06) (ii) Define injective and bijective function with example?(04)	10
Q 4(a)	(i) Find the minimum spanning tree (using Kruskal's Algorithm). (ii) What are the different conditions for the graphs to be isomorphic ?	10



Q 4 (b)	(i) What are the different conditions for Hamiltonian path and circuit. Justify with example. (ii) What are the different conditions for Euler path and circuit. Justify with example.	10
Q5 (a)	Prove that the set $G = \{ 1,2,3,4,5,6 \}$ is a finite abelian group of order 6 with respect to multiplication modulo 7.	10
Q 5 (b)	a) Consider the $(3, 8)$ encoding function $e : B^3 \rightarrow B^8$ as $e(000) = 00000000$ $e(100) = 10100100$ $e(001) = 10111100$ $e(101) = 10001001$ $e(010) = 00101101$ $e(110) = 00011100$ $e(011) = 10010101$ $e(111) = 00110001$ Find out how many errors will it detect and correct? Find all the possible combinations.	10
OR		
Q5(b)	a) Let $H =$ $0 \ 1 \ 1$ $1 \ 0 \ 1$ $1 \ 0 \ 0$ $0 \ 1 \ 0$ $0 \ 0 \ 1$ Be a parity check matrix, Determine the group code $e_H : B^3 \rightarrow B^6$	10